Opposition report for degree project

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DV1478: BACHELOR THESIS IN COMPUTER SCIENCE

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Thesis	Title	Particle-Oriented Bounding Box Skeletal Animation (POBBSA)
	Author(s)	Simon Strand and Karlos Napa Häger

1 Introduction

This study discusses a technique used to animate objects that are made from loose materials such as sand or water. The two authors, Simon Strand and Karlos Napa Häger accomplish this by allowing billboarded particles to move with the model using oriented bounding boxes (OBB) to define their behavior. This opposition is conducted by reading the thesis rapport and taking notes on the parts considered important or problematic. The grammar and spell checking was done with the help of Grammarly as it is useful for checking large sections of the text. Additionally, the problematic structure of the text is noted when reading through the rapport as Grammarly does not show these types of flaws.

The rapport does a good job of explaining some topics, however, there are some sections where the descriptions are hard to follow as concepts are not always explained. In general, the rapport is well-written and informative but suffers from problematic sentences that are long and hard to understand. The rapport is structured with many nestled sub-sections which makes it hard to follow along.

2 Critical review

2.1 Abstract

The abstract in its current form is quite lengthy, almost an entire page. The part where the results are described is very short, only one sentence, and the information given is very superficial and does not match the results. Some of the abbreviations used are not explained such as "OBB" and "FPS", the latter of which is more problematic as it can have different meanings such as "frames per second" or "first-person shooter". Furthermore, the explanation of skeleton animation is incorrect, bones in skeleton animations do not carry weights only the vertices, however, later in the thesis it is explained correctly.

2.2 Introduction

Most parts of the Introduction are well-written and infer to the reader why the subject in this thesis is important. The background subsection is also informative, describing how normal animations are done and how they differ from voxel and particle-based animations. The rest of the introduction only has a few problems, most of these are grammatical errors, but, the information is good.

Aims and objectives include some parts with explanations that fit better in methodology as it explains how particle-oriented bounding box skeletal animation (POBBSA) works, however, it is only a small part so it is not distracting. The rest of these sections and sub-sections does a good job of explaining what will be tested in this thesis and what limits have been put in place. The research questions seem logical for what wants to be studied. Although, they are not motivated or explained in the sub-section which would give the reader a deeper understanding of why the two were chosen. There is no section with an outline that could further help the reader understand what the thesis will discuss and where those parts are.

2.3 Related Work

The related work section is short, five scientific papers are given as related work. Two of these are on hitboxes and one of them is on deforming meshes which do not relate to the technique implemented in this study. Moreover, these two are only briefly described and it is not stated why these two are relevant. The other three sources are relevant to the text and are described well. Still, more of a reason there where chosen and how they relate can be explained. There is no section on Gap in related work.

2.4 Method

The method suffers from multiple problems some of which make this rapport very hard to read. The excessive use of deep sub-sections and the constant reference to other sub-sections means a very fragmented reading experience. It is not helped by the authors choice to only refer to the section by the numbers alone, such as "(read section 3.2.1.4 for more info)". The structure of the method is such that the references mostly direct the reader to a chapter that is ahead and has not been read yet. Some of the sub-chapters are very small and only contain a few sentences leading to an excessive number of headers, resulting in a less coherent section. In many parts of the text, information is given in parenthesis, at times entire sentences. This text should be incorporated into the text to increase the flow for the reader.

Figures are used in this chapter to show the models, among other relevant elements. Out of all the pictures used only one is mentioned in the text while the others are never mentioned. Images are meant to be used in a way to help the reader, not only for decorative purposes. The images should be tied to the text to give the reader more information or visually explain concepts. The captions of the images are descriptive which gives the reader an understanding of what the images show, tying the images to the text the reader can gain a more comprehensive understanding of the subject matter.

The description of the behavior of the particles in section "3.2.3.2 Style" are well described and gives the reader a point list of how they behave during all parts of the animation. However, the descriptors could benefit from minor restructuring to improve their clarity and make them easier to understand. In that vein, some words are repeated many times during the two descriptions which makes them sound repetitive.

The section on performance testing do a good job of describing what will be tested in the experiments. The first experiment is on the performance of the implementation where the system under test (SUT) is described and the methodology is described. Frame rate and memory usage were monitored to compare POBBSA to traditional skeleton animation at different numbers of particles. The tests were done over a time of five minutes per test. The information given on the SUT is too specific and not interesting, the reader is not interested in the specific make and model of the GPU used in the tests. The section on the survey where the quality of POBBSA will be judged is also well described. The important questions are listed with an explanation of why they are important. However, there is no chapter on what has been done to get more reliable data or why this data is trustworthy. Furthermore, there is no discussion of why the chosen research methodology is the most suited for the given task. Lastly, it is never mentioned how sensitive data will be handled in accordance with GDPR.

2.5 Result and Analysis

Early in this chapter the number of participants should be mentioned so the reader knows when reading the graphs. The graphs are very cluttered and crammed as a whole. Titles are missing, instead, a combination of the vertical axis and the horizontal axis is utilized as titles to describe what the graphs are showing, this makes the graphs hard to tell apart. Additionally, the graphs are organized in such a way that the vertical axis titles are almost intersection one another making them hard to read. Furthermore, the positioning of the graphs on the right side stretches out on the margins of the document. Lastly, the legend is close to the columns in the graph which further decreases the readability. The charts display the number above the columns which makes them easy to compare to one another.

In the next chapter, the same tests are done for another more complex model and the results are visualized in the same format as before. Usually, this is ideal as the reader has been familiarized with the way the graphs are displayed. However, the results in this chapter suffer from the same issues as described in the previous paragraph.

When presenting the survey results, all of the graphs have captions that help the reader understand what the graphs show. Still, the graphs do not have titles making it necessary for the reader to reference the captions. The numbers above the bars are missing from these resulting in making them hard to compare. In the text, the graphs are referred to as Q0 - Q7 while only one of the graphs has that name in the caption whereas the others are called "Figure 4.4" - "Figure 4.10". For the six who are captioned as figures, they have the Q name on their horizontal axis, and the information that is normally there is in a box below the graph where the colours match them to their bar. None of the vertical axis titles are very descriptive.

2.6 Discussion

The discussion mainly focuses on the survey that where conducted while the results of the performance testing are only briefly discussed. An explanation is given as to why the results look the way they do, the lower memory consumption and lower framerate are well explained. However, there is no discussion regarding the performance.

The survey showed that one of the models was preferred with POBBSA and the other with traditional skeleton animations. This is discussed and explained to be the result of the complexity difference in the models where POBBSA struggles with high complexity. It is demonstrated that the participants only noticed a small difference in animation when the simple Minecraft model was used and a large difference when the complex Stormtrooper was used. POBBSA was preferred on the Minecraft model while traditional animations for the Stormtrooper. The authors claim that POBBSA has a future in media, however, it is not backed up by their data gathered from the survey. The authors also discuss that the technique has a preference for cuboid characters which would make it difficult to say that the technique has a future in media.

2.7 Future Work and conclusion

The future work is well thought out with relevant ways to improve POBBSA and further test its limits. A total of four future studies are suggested two of which are acceleration techniques and the other two are to visually improve POBBSA. The first example given would be to test how different behaviors of the particles would change the final look of the animations. The other suggestion that can improve flexibility and visuals is to try different behavior, again.

Another suggested future work is to try using different shapes in addition to making collisions, such as bounding spheres and Axis-aligned bounding boxes. This could speed up the collision checks and potentially help lower the performance difference between POBBSA and skeleton animation. However, no data is provided regarding the computational time for collision checks in this study, which could influence the feasibility and relevance of conducting such a follow-up study.

Finally, the conclusion wraps up the rapport by giving a brief summary of the results and the conclusion.

2.8 Other Thoughts

In chapter 3.2.1.3 it is mentioned that the vertices of the mesh are converted to particles that then behave as defined by the styles, from this, it appears that the original mesh is not saved. This is further backed by the fact that POBBSA uses less memory than traditional animation. Both styles state that if a particle falls outside an OBB, it will respawn within a random OBB after traveling a certain distance. This should mean that models can lose details by shedding particles in certain parts of the models that move a lot. However, this could be mitigated by having a lot of particles which could help short term, still, the longer a character exists the more particles should be trapped in bones that do not drop many particles such as the chest. Furthermore, the particles do not interact with each other so they will not push each other out of the chest meaning only very rapid movement of the whole body will shed these particles. This could result that the parts of a mesh that move a lot, such as hands, would lose a lot of their particles and stationary bones gaining more. At first, this would result in holes in the model while after some time parts of the body might go missing. The result could be hands without any particles, yet still gripping weapons or other objects, creating an illusion of floating items instead of them being held. The artifacts the authors mention are probably an early sign of the decreased distribution of particles.

3 Required changes

This rapport needs some changes before it can be published. In this chapter, some changes that need to be remedied will be presented.

3.1 Missing sections

The report should include a section on the research gap. This section will highlight the lack of previous studies on the topic or the unexplored areas that this study aims to address. It will also establish the relevance of this study and its potential contribution to the scientific world.

The addition of a "Validity and reliability of your approach" section would increase the readers confidence in the results. This also adds an opportunity where the authors can give arguments for why the methodology is appropriate.

3.2 Structual

The method's structure needs to be reorganized to improve readability. Currently, it is quite challenging to follow. Some sections, like "3.2.1.1 Load Colors," are small and might be unnecessary. Moreover, the deep sub-chapter format used is not included in the table of contents. As a result of the sub-chapters, it becomes difficult to understand references to other chapters. For example, the reader may struggle to recall what subchapter 3.2.1.2 contains when it is mentioned as "previously described in section 3.2.1.2."

To enhance readability, the text should be written in a way that presents information before it is required. This approach allows the reader to grasp the necessary details without having to pause and read other chapters ahead of the current one. Additionally, the texts that are written in parentheses should be incorporated into the text, an example can be found in "3.2.2.1 Update animator", there is more than just this one in the texts. Some of the explanations are complicated and need to be rewritten. An example is the first sentence in "3.2.2.2 Update bone transformation", again, this is one of many.

3.3 Data Visualization

The first set of graphs shows the results from the performance testing that need minor adjustments. The text on the axis should display what is on that axis, such as FPS on the vertical, and technique used on the horizontal axis. The graphs should also have descriptive titles that tell the user what the graph is showing without needing to read the caption or the text. The graphs are positioned in such a where two are close to intersecting leading to less clarity of what text belongs to what graphs.

The graphs for the user study are different from the previous ones. These graphs do not show the number of responses above the pillars like the others did, this makes them hard to compare to one another, this should be added. These figures have captions unlike the previous but as mentioned before the names they are referred to in the text should be the same as the cation. Similar to the other graphs, the names on the axis need to be more descriptive. Additionally, the graphs should be as similar as possible as the reader is already familiar with the format from previous.

3.4 Discussion

The discussion section of the report is small and there are no parts that discuss the implication the results can have. This is one of the most important chapters where the authors can show that they understand the implications and draw conclusions based on their results. This needs to be added to the report before it can be published. A potential discussion topic could be where POBBSA works well, such as with simple character designs where the particle effects can compensate for the simplicity.

3.5 Particle distrobution

As described in section 2.8 Other Thoughts the potential particle distribution should be addressed in a section in Discussion. This will increase trustworthiness by showing the readers that the technique is not flawless. Moreover, this will also fit in with one of the suggested future works that address the artifacts.

4 Recommended changes

Here are minor changes that should be changed but are not required. Implementing these changes would make the rapport a lot easier to read. These will be in a point list as there are quite a few of them and they are self-explanatory.

- The excessive use of the word author could be changed by removing the actor from the text. An example "In this project, the authors made the origin of the OBB bones at (0.5, 0, 0.5)" could be changed to "In this project, the origin of the OBB bones was placed at (0.5, 0, 0.5)".
- Some of the related work can be explained more why it is related to this study and how it affected design choices.
- A flow chat to explain the user study can help the reader understand how it was conducted.
- Spell checking needs to be done, such as changing billboard-ed to billboarded.
- Make the description of how skeleton animation works consistent, and correct, in the text.
- Shorten the abstract while better representing the findings when describing the results.
- Explain the abbreviations when first using them. For example, FPS and OBB in abstract.
- Remove unnecessary information on the hardware used for the SUT, as mentioned previously. Calling the GPU a GTX 1070 gives enough context, it is not necessary to say that it is made by ASUS.
- Refer to image 4.2 in section 4.1.1 to show the artifacts visually and not just describe them in text.
- Remove ambiguity from the text, such as "small things are different" in "3.2.1 Preprocess" and "force changes from low to ultra" in "4.2 Quality". These terms could be explained to the reader so they do not need to assume their meaning.
- Add more related work and describe how they are relevant to the study.
- The title of the paper could be more descriptive. For example including that the technique is real-time
- Add a section on data security and what sensitive data will be gathered.
- Make the two future works on trying different styles to one future work to avoid repetition.
- Change the keywords so that not all of them are in the title.